

Introduction to Information Security

1. Define Information. Explain the components of an Information System.
 2. Define information security and explain its primary objectives.
 3. Define information security and briefly describe its domains.
 4. What is the CIA triad in information security? Describe each component with an example.
 5. How does the Parkerian hexad extend the CIA triad? List the additional principles and their significance.
 6. Explain Eugene Spafford's statement about the "only truly secure system." What does it imply about the balance between security and usability?
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Information Security Attacks

7. Define attacks in the context of information security.
 8. Write a short note on the types of attacks.
 9. Write a short note on the risk management model.
 10. Write a short note on Defense in Depth.
 11. Differentiate between physical and logical assets in the context of information security.
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Crypto Basics

6. What is Kerckhoffs' Principle? Why is it important in cryptographic system design?
7. Describe the basic process of encryption and decryption in a cryptosystem.
8. Explain the difference between symmetric and public key cryptosystems.
9. What is a Caesar cipher? Describe its encryption and decryption processes.
10. Discuss the limitations of the Caesar cipher and how it can be attacked.
11. A plaintext message "HELLO" is encrypted using a Caesar cipher with a shift of 3. Write the ciphertext.
12. A Caesar cipher with a shift of $n=4$ produces the ciphertext "CSYEVIIVQMREXIH". Decrypt this ciphertext to recover the plaintext.

13. What is the total number of possible keys in a Caesar cipher? Show the calculation.
14. Explain the difference between symmetric and public key cryptosystems.
15. Encrypt the message “welcome to cse ju” using a double transposition cipher with 4 rows and 4 columns, using the row permutation $(1,2,3,4) \rightarrow (2,4,1,3)$ and the column permutation $(1,2,3,4) \rightarrow (3,1,2,4)$.
16. Suppose you are given a one-time pad = 111 101 110 101 111 100 and alphabet encodings,

ALPHABET **E** **H** **I** **K** **L** **R** **S** **T**

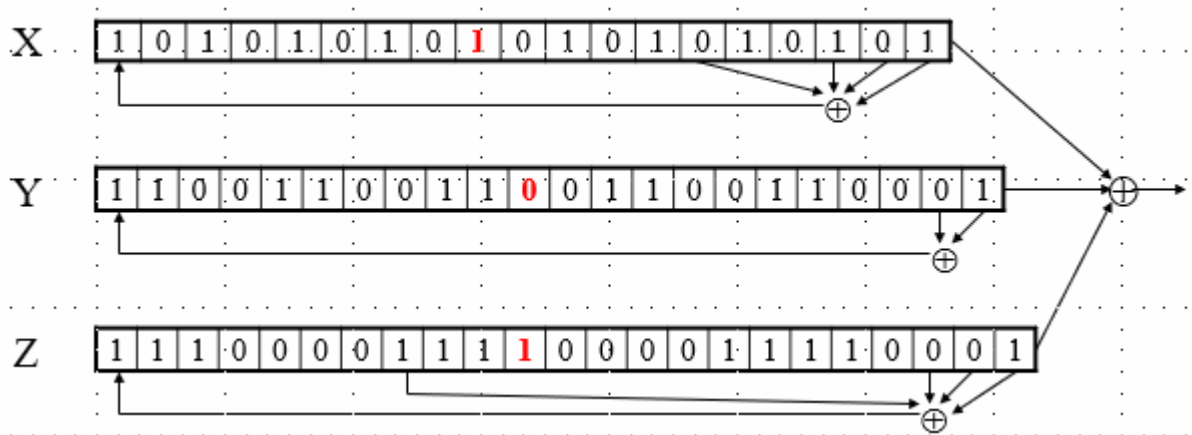
BINARY	000	001	010	011	100	101	110	111
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Now determine the ciphertext for the following plaintexts

- i. THRILL
- ii. TILLER

Crypto Part II

17. What is a stream cipher? Provide examples of situations where it is used.
18. Describe the structure and operation of the DES algorithm, including its key size and block size.
19. What role do S-boxes play in block ciphers like DES?
20. Explain the difference between stream ciphers and block ciphers.
21. How does the A5/1 stream cipher generate its keystream? Include a brief explanation of its shift registers.
22. The A5/1 cipher uses three shift registers X,Y,Z with lengths of 19, 22, and 23 bits, respectively.



Suppose the majority function mmm at a specific time is computed as:

$m = \text{maj}(X[8], Y[10], Z[10])$

Given the initial states: $X[8]=1$, $Y[10]=0$, $Z[10]=1$. Compute the value of m , determine which registers will step, and calculate the new state of X if its feedback polynomial is:

$t = X[13] \oplus X[16] \oplus X[17] \oplus X[18]$

23. If we generate a sufficiently long keystream, the keystream must eventually repeat. Why?
24. Why is it a security concern if the keystream repeats?
25. Answer the following questions
 - a. On average, how often does the X register step?
 - b. On average, how often does the Y register step?
 - c. On average, how often does the Z register step?
 - d. On average, how often do all three registers step?
 - e. On average, how often do exactly two registers step?
 - f. On average, how often does exactly one register step?

Crypto Part III

26. What improvements does AES offer over DES? List its key features and block size.
27. Suppose you have the following S-block of a DES model.

Column →	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Row 0	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
Row 1	0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
Row 2	4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
Row 3	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

Now, for the input sequence 101101, answer the following.

- i. Which row contains the output value?
- ii. Which column contains the output value?
- iii. What will be the value of the output in decimal?
- iv. What will be the value of the output in binary?

28. Describe the four functions used in an AES encryption round.

29. Compare ECB (Electronic Codebook) and CBC (Cipher Block Chaining) modes of operation. Highlight their advantages and disadvantages.
30. What is the purpose of an initialization vector (IV) in CBC mode?
31. Explain the concept of a Message Authentication Code (MAC). How does it ensure data integrity?
32. A digital signature provides for data integrity and a MAC provides for data integrity. Why does a signature also provides for non-repudiation while a MAC does not?
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General Cryptography

33. RSA Key Generation

Let $p=11$ and $q=13$. Compute the modulus N , Euler's totient $\phi(N)$, and find a valid public key exponent e such that $\gcd(e, \phi(N))=1$.

34. RSA Encryption

Using the RSA public key ($N=143, e=7$), encrypt the message $M=10$. Show all steps.

35. RSA Decryption

Using the private key $d=103$ and modulus $N=143$, decrypt the ciphertext $C=100$. Show the complete calculation.

36. Diffie-Hellman Key Exchange

Alice and Bob use the Diffie-Hellman key exchange with $p=23$ and $g=5$. If Alice selects her private key $a=6$ and Bob selects $b=15$, compute the shared secret key K .

37. Knapsack Encryption

Given the superincreasing knapsack $(2,3,7,14)$, a modulus $n=31$, and a multiplier $m=5$, compute the general knapsack.

38. Knapsack Decryption

Using the general knapsack $(10,15,5,8)$ and ciphertext $C=23$, decrypt the message to retrieve the binary sequence.

39. ECC

A digital signature provides for data integrity and a MAC provides for data integrity. Why does a digital signature also provides for non-repudiation while a MAC does not? Consider the elliptic curve $E : y^2 = x^3 + 7x + b \pmod{11}$.

- Determine b so that the point $P = (4,5)$ is on the curve E .
 - Using the b found in part a, list all points on E .
 - Using the b found in part a, find the sum $(4,5) + (5,4)$ on E .
 - Using the b found in part a, find the point $3(4,5)$.
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Hashing

40. What are the properties of a Cryptographic Hash Function?
 41. State the applications of Hash function.
 42. Draw the block diagram of SHA-3.
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Access Control

43. Evaluate the limitations of traditional password-based authentication in terms of human usability, scalability, and vulnerability to attacks.
44. Assess the use of biometrics for authentication. What trade-offs exist between user convenience, error rates, and security?
45. Describe how a salted hash strengthens password security and explain how it mitigates precomputed dictionary attacks.
46. Discuss the advantages and risks associated with single sign-on (SSO) systems. How do they improve user experience, and what vulnerabilities do they introduce?
47. In the context of biometrics, define the terms fraud rate and insult rate. In statistics, which of these is a Type I error and which is a Type II error?
48. Suppose all passwords on a given system are 8 characters and that each character can be any one of 64 different values. The passwords are hashed (with a salt) and stored in a password file. Now suppose Trudy has a password cracking program that can test 64 passwords per second. Trudy also has a dictionary of 230 common passwords and the probability that any given password is in her dictionary is $1/4$. The password file on this system contains 256 password hashes.
 - a. How many different passwords are possible?
 - b. How long, on average, will it take Trudy to crack the administrator's password?
 - c. What is the probability that at least one of the 256 passwords in the password file is in the dictionary?
 - d. What is the expected work for Trudy to recover any one of the passwords in the password file?